

CREST

Criteria Retrieval, Extraction & Selection Tool

An AI-Assisted Decision Support System for Criteria Selection in Multi-Criteria Decision Making

User Manual v.1

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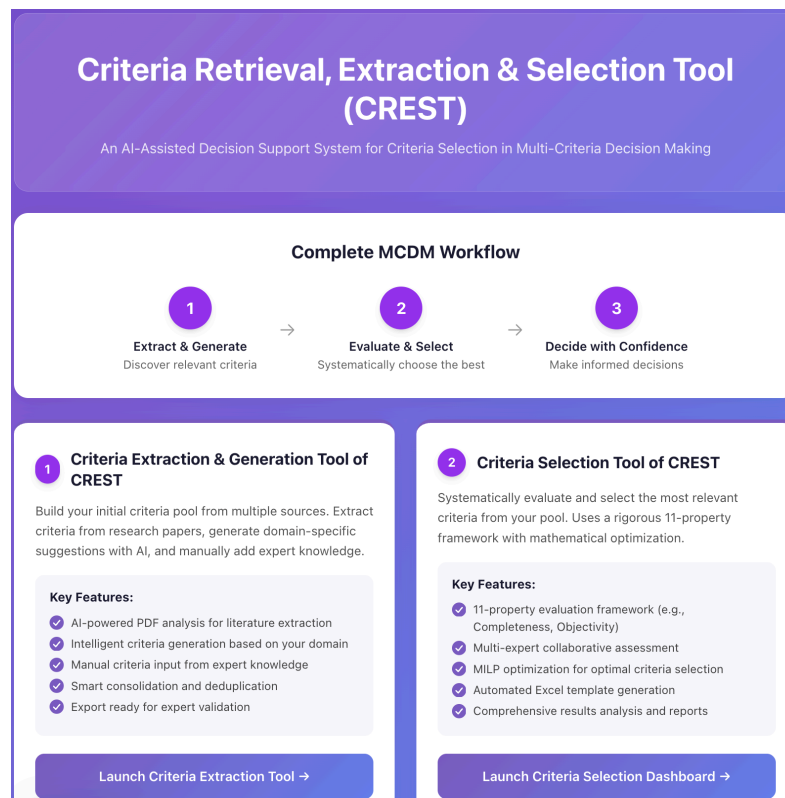
Overview

The **Criteria Retrieval, Extraction and Selection Tool (CREST)** is a comprehensive web-based decision support system designed to help researchers and decision-makers systematically identify, extract, generate, and select the most appropriate evaluation criteria for multi-criteria decision making (MCDM). Main page: [CREST main page](#)

CREST consists of two integrated phases:

Phase 1 — Criteria Extraction & Generation Dashboard. Helps you build a comprehensive pool of potential criteria from multiple sources: expert knowledge, AI-powered generation, and academic literature analysis. [Open Phase 1](#)

Phase 2 — Criteria Selection Dashboard. Uses a rigorous 11-property evaluation framework with mathematical optimization to systematically select the best criteria subset from your pool. [Open Phase 2](#)



Complete Workflow Overview

CREST guides you through a comprehensive 10-step process divided into two phases. This manual provides step-by-step instructions with visual references for each phase of the process.

Phase 1: Criteria Extraction & Generation (Steps 1–6)

This module helps you build a comprehensive pool of evaluation criteria using Ralph Keeney's Value-Focused Thinking framework (Keeney, 1996). You'll integrate criteria from three sources: manual expert input, AI-powered generation, and literature analysis. The following sections provide an overview of the steps in Phase 1, then break down each step in detail.

Step 1: Claude API Setup (Required)

Claude-Powered MCDM System

Phase 1: Criteria Extraction & Generation Tool of CREST

What This Tool Does:
This system helps you create comprehensive criteria sets for multi-criteria decision making by:

- **Analyzing research papers:** Automatically extracts evaluation criteria from uploaded PDF documents using Claude AI
- **Generating domain-specific criteria:** Creates relevant criteria based on your problem description
- **Manual input:** Allows you to add known criteria and expert knowledge
- **Smart consolidation:** Merges and deduplicates criteria from all sources
- **Export for validation:** Provides structured outputs for expert review and validation

What You'll Do:
Required steps: 1 → 2 | **Optional steps:** 3, 4, 5 | **Final step:** 6
Follow the numbered tabs below in sequence. At least one optional step (3, 4, or 5) should be completed before generating results.

Academic Transparency:
All AI-generated criteria are traceable. The system is designed for academic research with full transparency about methodologies and sources.

1. Claude Setup (Required)

2. Domain Setup (Required)

3. Manual Addition (Optional)

4. AI Criteria Generation (Optional)

5. PDF Analysis (Optional)

6. Consolidate & Export

1. Configure Claude AI (Required)

What's Happening Behind the Scenes?
This system uses Claude AI (by Anthropic) to analyze PDF documents, generate domain-specific criteria, and provide source transparency. Your documents are processed securely — no data is stored permanently.

Option 1: Use Provided Research API Key (Recommended – Free for Academic Use)

For Academic Research – No Setup Required! Completely free, instant access, no registration needed.

Research API Key (Pre-filled)

.....

Claude Model

Claude 3.5 Sonnet (Recommended for Research)

Use Research API Key

Option 2: Use Your Own Anthropic Claude API Key (Advanced)

Next Phase: 2. Domain Setup →

This step configures the AI engine that powers automated criteria extraction and generation.

Instructions:

1. **The research API key is pre-configured for academic use.**
2. Select your preferred Claude model:
 - **Claude 3.5 Sonnet** (recommended for research)
 - Claude 3.5 Haiku (faster analysis)
 - Claude 3 Opus (most capable)
3. Click "Use Research API Key".
4. Verify the success message appears.
5. Click "Next Phase: 2. Domain Setup" to proceed.

Note: For personal API usage, expand "Option 2" and enter your own Anthropic API key from console.anthropic.com.

Step 2: Domain Setup (Required)

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(Required)

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Generation
(Optional)

5. PDF Analysis
(Optional)

6. Consolidate &
Export

2. Define Your Decision Domain (Required)

Describe your specific decision-making problem. This information helps the AI generate relevant criteria tailored to your context.

Domain Name

Energy

Detailed Context

Describe your situation, constraints, objectives, and requirements...

Decision Objective

Environmental Performance (O1); Cost Efficiency (O2); Deployment & Operations (O3); Social Acceptance & Engagement (O4); Reliability & Resilienc

Save Domain Setup

Warning: Please save your domain setup before proceeding to other steps.

← Back: 1. Claude Setup

Next: 3. Manual Addition →

Next: 4. AI Generation →

Next: 5. PDF Analysis →

Define your decision-making problem context to guide criteria generation.

Instructions:

6. **Domain Name:** enter a descriptive name (e.g., "Renewable Energy Technology Selection").
7. **Detailed Context:** describe your situation, constraints, and requirements (2–4 sentences).
8. **Decision Objectives:** list your fundamental objectives, one per line or separated by semicolons.
 - Example: "Minimize environmental impact; Maximize cost efficiency; Ensure system reliability"
9. Click "Save Domain Setup".
10. Review the summary confirmation.
11. Proceed to at least one of the optional steps (3, 4, or 5) before going to Step 6.

Important: Decision objectives are used to categorize all generated criteria, so be specific.

Step 3: Manual Addition of Known Criteria (Optional)

Claude-Powered MCDM System

Phase 1: Criteria Extraction & Generation Tool of CREST

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1. Claude Setup
(Required)

2. Domain Setup
(Required)

3. Manual Addition
(Optional)

4. AI Criteria
Generation
(Optional)

5. PDF Analysis
(Optional)

6. Consolidate &
Export

3. Manual Addition of Known Criteria (Optional)

Add criteria you already know are important — from regulations, policies, expert knowledge, or prior experience. Add them one by one.

Criterion Name

Description

Add Criterion Clear All

No criteria added yet. Add your first criterion above.

← Back: 2. Domain Setup Next: 4. AI Generation → Next: 5. PDF Analysis → Skip to 6. Results →

Add evaluation criteria you already know are important based on expert knowledge, regulations, or previous experience.

Instructions:

12. **Criterion Name:** e.g., "Maintenance Requirements".
13. **Description:** explain what this criterion measures.
14. Click "Add Criterion".
15. Repeat for each criterion you want to add manually.
16. Review the criteria list below the form.
17. Use the "X" button to remove any unwanted criteria.
18. Click "Clear All" to start over if needed.

Example use cases:

- Adding criteria required by industry standards
- Including factors from regulatory frameworks
- Incorporating expert-identified evaluation factors

Step 4: AI Criteria Generation (Optional)

1. Claude Setup
(Required)

2. Domain Setup
(Required)

3. Manual Addition
(Optional)

4. AI Criteria
Generation
(Optional)

5. PDF Analysis
(Optional)

6. Consolidate &
Export

4. Generate Criteria with Claude AI (Optional)

Let Claude analyze your domain and generate intelligent criteria suggestions based on academic research and best practices.

Claude content

How AI Generation Works:
Claude uses its training on academic literature and decision-making frameworks to suggest relevant criteria for your specific domain, considering your context and objectives.

Additional Instructions for Claude
Examples: "Must comply with GDPR" | "Budget under \$50k" | "Focus on scalability" | "Emphasize security"

Suggest the user-specified number of the most important evaluation criteria for selecting the best energy procurement option for a hypothetical data center, with emphasis on high-impact criteria that are often overlooked or insufficiently considered in practice. Focus on criteria that are relevant, measurable, and useful for comparing renewable and conventional energy alternatives under reliability and sustainability objectives.

Number of Criteria: 7

Generate with Claude AI

Carbon Emissions Intensity with Additionality Adjustment

Lifecycle CO₂-equivalent emissions per MWh (gCO₂e/MWh), adjusted downward only for renewable attributes that meet strict additionality criteria: the generation asset must be (1) new (≤5 years old), (2) located in the same grid balancing area, and (3) temporally matched on an hourly basis rather than annualized. Unadjusted RECs or annual matching are scored without the additionality discount.
Category: Environmental Sustainability
Reasoning: Annual REC matching—the dominant industry practice—allows data centers to claim net-zero while drawing fossil power hour-by-hour. Hourly carbon accounting (championed by Google and the UN 24/7 CFE initiative) is the emerging gold standard but remains widely underweighted in procurement scorecards. This criterion penalizes greenwashing and rewards genuinely additional clean electrons, which is critical for Scope 2 reporting credibility under GHG Protocol Market-Based and Location-Based methods.
Frameworks: GHG Protocol Scope 2 Guidance, UN 24/7 Carbon-Free Energy Compact, SBTi Corporate Net-Zero Standard, EnergyTag Granular Certificate standard

Claude AI

Supply Reliability and N-1 Contingency Coverage

The probability that contracted power delivery meets or exceeds 99.99% uptime (≤52.6 min/year interruption), measured as the percentage of contracted capacity still deliverable under an N-1 transmission or generation contingency scenario. Quantified via probabilistic reliability studies (Loss of Load Probability, LOLP) at the specific interconnection point.
Category: Operational Reliability
Reasoning: Data centers are Tier III/IV facilities where a single unplanned outage can cost millions per minute. Procurement decisions frequently evaluate price and carbon credentials while treating grid reliability as a binary yes/no checkbox. LOLP-based contingency scoring forces suppliers to demonstrate redundancy depth, distinguishing contracts backed by firm transmission rights from interruptible or curtailable service—a critical but often invisible distinction.
Frameworks: IEEE 493 (Gold Book) reliability standards, NERC Reliability Standards (TPL-001), ANSI/TIA-942 data center tier classification, PJM/MISO LOLP methodologies

Claude AI

← Back: 3. Manual Addition

Next: 5. PDF Analysis →

Skip to 6. Results →

Generate domain-specific criteria using Claude AI based on your problem description.

Instructions:

- Additional Instructions:** enter any specific requirements or focus areas (optional).
- Generation Focus:** select the analysis emphasis (Comprehensive, Financial, Technical, User-Centric, or Risk-Based).
- Number of Criteria:** use the slider to select 5–20 criteria (default: 10).
- Click "Generate with Claude AI".
- Wait 10–30 seconds for AI analysis.
- Review generated criteria with their linked objectives and justifications.
- Remove unwanted criteria using the "X" button.

Output: Each AI-generated criterion includes a name and description, a linked objective from your Step 2 input, and a justification based on Keeney's framework.

Step 5: PDF Analysis & Literature Extraction (Optional)

1. Claude Setup
(Required)

2. Domain Setup
(Required)

3. Manual Addition
(Optional)

4. AI Criteria
Generation
(Optional)

5. PDF Analysis
(Optional)

6. Consolidate &
Export

5. Extract Criteria from Research Papers with Claude AI (Optional)

Upload research papers and let Claude extract relevant criteria using advanced reasoning and literature analysis.

Academic Transparency & Source Traceability:

- **Text extraction:** System extracts readable text from PDF documents
- **Claude analysis:** AI identifies evaluation frameworks, criteria lists, and decision factors in the literature
- **Source attribution:** Each extracted criterion is linked to its source document

Note: Works best with text-based PDFs. Scanned documents may require manual review.

Upload Research Papers

Choose Files 4 files

Analysis Focus

What should Claude focus on? e.g., 'Look for evaluation criteria for software selection'

Maximum Criteria per PDF: 10

Analyze with Claude AI

Net Present Cost (NPC)

Total present value of all costs associated with installing and operating a hybrid energy system over its lifetime, including initial, replacement, O&M, and fuel costs minus salvage value.
Category: Economic | Source: 1-paper.pdf

PDF Extract

Cost of Electricity (COE)

The average cost per unit of electrical energy produced by the hybrid energy system, used to assess economic viability and affordability of power supply.
Category: Economic | Source: 1-paper.pdf

PDF Extract

Loss of Power Supply Probability (LPSP)

A reliability indicator representing the probability that the energy system fails to meet the required load demand during a given period.
Category: Technical | Source: 1-paper.pdf

PDF Extract

Energy Curtailment Rate (ECR)

The fraction of generated renewable energy that cannot be used or stored and is therefore wasted, indicating system efficiency and sizing appropriateness.
Category: Technical | Source: 1-paper.pdf

PDF Extract

Extracted Text Preview

power and other energy utility services to all is critical for sustainable development. Developing decentralized hybrid energy systems may be one of the potential options for better energy sustainability. Determination of sustainability factors of such new technologies is critical in this regard. These factors may not converge to a single optimal solution for the best sustainability. Multi-criteria decision-making methods are used as the tool to decide an overall optimum sustainable solution based on these sustainability factors. This study presents a comprehensive review of published studies on different decision-making processes for sustainable decentralized hybrid energy systems. The probable future path of research is also discussed in this section. *Nonlinearities Abbreviations: AHP Analytic Hierarchy Process; ANP Analytic Network Process; CPS Feedback*

Analysis Log

Ready for PDF analysis... [18:42:45 PM] Processing 3/4: 1-paper.pdf [18:42:56 PM] Extracted 10 criteria from 1-paper.pdf [18:42:56 PM] Processing 2/4: 2-paper.pdf [18:43:07 PM] Extracted 10 criteria from 2-paper.pdf [18:43:07 PM] Processing 3/4: 3-paper.pdf [18:43:15 PM] Extracted 8 criteria from 3-paper.pdf [18:43:15 PM] Processing 4/4: 4-paper.pdf [18:43:28 PM] Extracted 5 criteria from 4-paper.pdf

← Back: 4. AI Generation

Next: 6. Results & Export →

Extract criteria from research papers using Claude's document analysis capabilities.

Instructions:

26. Click "Choose Files" and select PDF research papers (multiple files allowed).
27. **Analysis Focus:** describe what Claude should look for (e.g., "Extract evaluation criteria for energy systems").
28. **Maximum Criteria:** use the slider to set 5–30 criteria per document (default: 15).
29. Click "Analyze with Claude AI".
30. Wait for text extraction and analysis (30–60 seconds per document).
31. Review the Extracted Text Preview to verify PDF processing.
32. Examine extracted criteria with source-file attribution.
33. Check the Analysis Log for technical details (optional).

What gets extracted:

- Evaluation criteria explicitly mentioned in papers
- Framework dimensions and assessment factors
- Performance metrics and indicators
- Each criterion linked to fundamental objectives

Step 6: Consolidate & Export Results

1. Claude Setup (Required) 2. Domain Setup (Required) 3. Manual Addition (Optional) 4. AI Criteria Generation (Optional) 5. PDF Analysis (Optional) 6. Consolidate & Export

6. Consolidate & Export Final Results

Review all criteria, uncheck any you want to exclude, then export your final selection.

39 Total Unique Criteria 0 Manual Input 7 AI Generated 33 Literature Extracted

Consolidate All Criteria

39 of 39 criteria selected for export Select All Deselect All

Export CSV Export JSON Preview Criteria Pool

Preview - 39 selected criteria: ID | Name | Category | Source

1	Lifecycle Carbon Intensity	Environmental	Claude AI 2
2	Levelized Cost of Energy	Cost Efficiency	Claude AI 3
3	Time-to-Operational Readiness	Deployment & Integration	Claude AI 4
4	Community Acceptance Index	Social Acceptance	Claude AI 5
5	Capacity Factor and Dispatchability	Reliability	Claude AI 6
6	Grid Integration Complexity	Integration	Claude AI 7
7	Regulatory Compliance Rating	Governance & Economics	PDF Extract 8
8	Net Present Cost (NPC)	Economic	PDF Extract 9
9	Cost of Electricity (COE)	Economic	PDF Extract 10
10	Loss of Power Supply Probability	Technical	PDF Extract 11
12	Loss of Power Supply Probability	Technical	PDF Extract 12
13	Loss of Power Supply Probability	Technical	PDF Extract 13
14	Loss of Power Supply Probability	Technical	PDF Extract 14
15	Loss of Power Supply Probability	Technical	PDF Extract 15
16	Loss of Power Supply Probability	Technical	PDF Extract 16
17	Loss of Power Supply Probability	Technical	PDF Extract 17
18	Loss of Power Supply Probability	Technical	PDF Extract 18
19	Loss of Power Supply Probability	Technical	PDF Extract 19
20	Loss of Power Supply Probability	Technical	PDF Extract 20
21	Loss of Power Supply Probability	Technical	PDF Extract 21
22	Loss of Power Supply Probability	Technical	PDF Extract 22
23	Loss of Power Supply Probability	Technical	PDF Extract 23
24	Loss of Power Supply Probability	Technical	PDF Extract 24
25	Loss of Power Supply Probability	Technical	PDF Extract 25
26	Loss of Power Supply Probability	Technical	PDF Extract 26
27	Loss of Power Supply Probability	Technical	PDF Extract 27
28	Loss of Power Supply Probability	Technical	PDF Extract 28
29	Loss of Power Supply Probability	Technical	PDF Extract 29
30	Loss of Power Supply Probability	Technical	PDF Extract 30
31	Loss of Power Supply Probability	Technical	PDF Extract 31
32	Loss of Power Supply Probability	Technical	PDF Extract 32
33	Loss of Power Supply Probability	Technical	PDF Extract 33
34	Loss of Power Supply Probability	Technical	PDF Extract 34
35	Loss of Power Supply Probability	Technical	PDF Extract 35
36	Loss of Power Supply Probability	Technical	PDF Extract 36
37	Loss of Power Supply Probability	Technical	PDF Extract 37
38	Loss of Power Supply Probability	Technical	PDF Extract 38
39	Loss of Power Supply Probability	Technical	PDF Extract 39

#36 Availability and Reliability of Electrical Power
The reliability and continuous availability of the electrical power network at the chosen location, critical for uninterrupted data center operation.
Category: Energy | Source: PDF Extract (4-paparc.pdf) PDF Extract

#37 Energy Consumption
The total energy consumed by the data center, including computing equipment and cooling systems, measured against sustainability targets.
Category: Energy | Source: PDF Extract (4-paparc.pdf) PDF Extract

#38 Cooling System Energy Requirements
The energy demand attributable to cooling infrastructure, which constitutes a significant portion of total data center energy usage.
Category: Energy | Source: PDF Extract (4-paparc.pdf) PDF Extract

#39 Proximity to Renewable or Low-Cost Energy Sources
The degree to which the location enables access to renewable or affordable energy sources to support sustainable and cost-effective data center operations.
Category: Energy | Source: PDF Extract (4-paparc.pdf) PDF Extract

Combine all criteria sources, select your final criteria set, and export it for validation.

Instructions:

34. Click "Consolidate All Criteria" to merge all sources.
35. Review the Statistics Dashboard showing breakdown by source.
36. Use checkboxes to select/deselect criteria for your final set.
37. Use "Select All" or "Deselect All" buttons for bulk operations.
38. Choose an export format:
 - **Export CSV:** data file for analysis tools.
 - **Export JSON:** structured format with full metadata.
 - **Generate Report:** comprehensive text report.
 - **Create Validation Form:** a structured template for expert review. This is the bridge between Phase 1 and Phase 2 — the validated criteria pool produced here becomes the input for Phase 2 optimization, so it is worth completing carefully.

Export contents:

- All selected criteria with descriptions
- Linked objectives and justifications
- Source attribution (Manual, AI, or PDF)
- Confidence scores and timestamps
- Framework metadata for academic transparency

Best practice: After export, use the validation form to gather expert feedback on criteria relevance and completeness before proceeding to Phase 2.

Phase 2: Criteria Selection Dashboard (Steps 1–4)

This module uses a rigorous 11-property evaluation framework combined with mathematical optimization to systematically select the best criteria subset from your pool. You'll collect expert judgments, configure decision parameters, and execute optimization. Step numbering restarts at 1 here because Phase 2 is a distinct dashboard with its own workflow.

Step 1: Generate Excel Template

Configure your problem parameters and generate a customized Excel template for data collection.

Instructions:

1. Enter problem structure (left column)

- **Number of Criteria:** total criteria you want to evaluate (e.g., 16).
- **Number of Alternatives:** options you're comparing (e.g., 7).
- **Number of Experts:** subject-matter experts providing input (e.g., 3).
- **Number of Objectives:** high-level goals for your decision (e.g., 7).

2. Set parsimony bounds (right column — top section)

- **Target Minimum (ω):** minimum criteria to select (e.g., 5).
- **Target Maximum (ζ):** maximum criteria to select (e.g., 9).
- These bounds ensure you get neither too few nor too many criteria.

3. Set objective representation bounds (right column — middle section)

- **Min representation per objective $L(o)$:** minimum criteria per objective (default: 1). Ensures every objective has at least this many criteria.
- **Max representation per objective $U(o)$:** maximum criteria per objective (default: 2). Prevents any single objective from being overrepresented.

4. Configure advanced thresholds (optional)

Click "⚙️ Advanced Thresholds" to expand. Default values shown:

- Completeness (α): 6.00
- Measurability Objective (γ_O): 6.50

- Measurability Subjective (γ_S): 5.50
- Distinctiveness (δ): 0.75
- Sensitivity (θ): 0.035
- Cost-Effectiveness Objective (τ_O): 7.00
- Cost-Effectiveness Subjective (τ_S): 6.00
- Alignment (λ): 6.50
- Cognitive Coherence (μ): 7.00

Leave defaults unless you have specific requirements.

5. Generate and download

- Click the red "📄 Generate Excel Template" button at the bottom.
- Download the generated file.
- Save it with a meaningful name (e.g., "MCDM_EnergySelection_Template.xlsx").

What happens next: The template contains 12 sheets where you collect expert judgments across 11 evaluation dimensions. Fill in all yellow cells only — green cells are auto-calculated.

Fill out the Excel template (before Step 2)

Sheet 0: Configuration

- **Criteria Names:** replace "Criterion 1" with descriptive names.
- **Criteria Types:** select "Cost" or "Benefit" from dropdown.
- **Alternative Names:** replace "Alternative 1" with actual option names.
- **Objective Names:** replace "Objective 1" with your strategic goals.

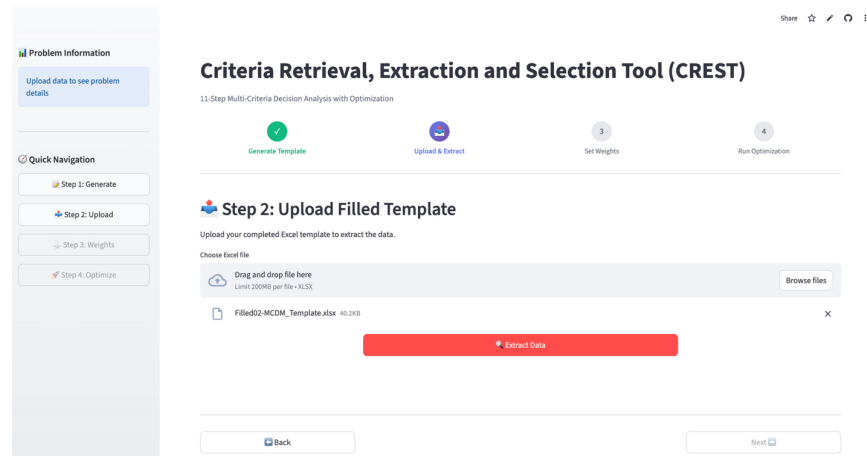
Sheets 1–11: Expert data collection

- **Sheet 1 (Completeness):** rate 1–10 how well each criterion covers the decision aspect.
- **Sheet 2 (Objectivity):** classify as objective (1) or subjective (0).
- **Sheet 3 (Measurability):** rate 1–10 how easily a criterion can be quantified.
- **Sheet 4 (Distinctiveness):** provide decision matrices (alternatives $\times$ criteria).
- **Sheet 6 (Sensitivity):** provide decision matrices for sensitivity analysis.
- **Sheet 7 (Cost-Effectiveness):** rate 0–10 resource efficiency of each criterion.
- **Sheet 8 (Alignment):** rate 1–10 how well criterion aligns with objectives.
- **Sheet 9 (Cognitive Coherence):** cross-expert ratings of definition clarity (1–10).
- **Sheet 10 (Monotone Coherence):** binary vote on monotonicity (1 or 0).
- **Sheet 11 (Representativeness):** assign criteria to objectives (1 = assigned, 0 = not).

Important: Fill only YELLOW cells. Do not modify green cells or formulas.

Step 2: Upload & Extract Data

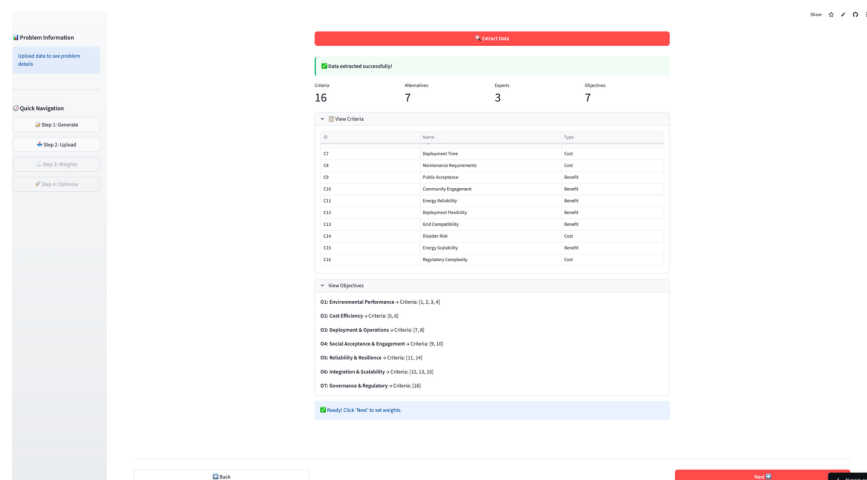
2.1 Upload your template



Instructions:

39. **Access Step 2 in the application.** The progress indicator at the top shows: Generate Template ✓ | Upload & Extract (active) | Set Weights | Run Optimization. You can also use the Quick Navigation sidebar to click "Step 2: Upload".
40. **Upload your completed template.** You will see "Step 2: Upload Filled Template". Click "Browse files" or drag and drop your Excel file. Limit is 200 MB per file (XLSX). The filename appears once selected (e.g., "Filled02-MCDM_Template.xlsx 40.2 KB"). You can remove the file with the X icon if needed.
41. **Extract the data.** Click the large red "Extract Data" button and wait 5–15 seconds while the system processes your file.

2.2 Review extracted data



After clicking "Extract Data", the page updates to show:

- **Success message:** Green box with checkmark "✓ Data extracted successfully!"
- **Problem summary metrics** (four columns): Criteria 16 | Alternatives 7 | Experts 3 | Objectives 7.
- **View Criteria** (expandable): full criteria table with ID, Name, and Type.

- **View Objectives** (expandable): which criteria are assigned to each objective.
- **Ready message:** "✅ Ready! Click 'Next' to set weights."
- **Navigation:** "⬅️ Back" and "Next ➡️" buttons.

Step 3: Set Weights (SWING Weighting)

The screenshot shows the 'Step 3: Swing Weighting' interface. The left sidebar contains navigation buttons for 'Criteria Selection', 'Step 3: Swing Weighting', 'Set Weights', and 'Run Optimization'. The main panel has a title 'Step 3: Swing Weighting' and a blue info box explaining the classical swing weighting method. Below the info box is a table with 12 criteria and their normalized weights. The right panel shows the 'Normalized Weights' table.

Criteria	Normalized Weights
1. Safety	0.0100
2. Reliability	0.0100
3. Durability	0.0100
4. Performance	0.0100
5. Cost	0.0100
6. Maintenance	0.0100
7. Fuel Economy	0.0100
8. Acceleration	0.0100
9. Braking	0.0100
10. Steering	0.0100
11. Visibility	0.0100
12. Noise	0.0100

This step uses a SWING-style weighting methodology (Keeney & Raiffa, 1976) to elicit the relative importance of different quality components. You'll identify which component has the most valuable swing from worst to best performance, then rate all other components relative to that reference.

Instructions:

1. Access the weighting interface

- Navigate to "Step 3: Weights" via progress indicator or sidebar.
- Progress shows: Generate Template ✓ | Upload & Extract ✓ | Set Weights (active) | Run Optimization.

2. Understand the interface layout

Left panel — Problem Information (sidebar):

- Summary metrics: Criteria 16 | Alternatives 7 | Experts 3 | Objectives 7.
- Quick Navigation buttons for all four steps.

Center panel — main interface:

- Title: "⚖️ Step 3: Swing Weighting".
- Blue info box explaining the classical swing weighting method:
 - Select the component with the most valuable swing from worst to best performance.
 - That component is assigned a reference value of 100.
 - Rate all other components relative to the reference (0.0 = no value, 100 = equally valuable).
 - Weights are automatically normalized to sum to 100.

3. Select reference component

Question: "Which component has the MOST valuable swing from worst to best performance?" Use the dropdown menu to select from 12 components. Each component represents a property used to evaluate your criteria set.

Benefit properties (higher is better):

- **Completeness (w_1)** — how comprehensively the criteria capture all relevant dimensions of the decision problem. Based on expert ratings (0–10 scale).
- **Objectivity (w_2)** — proportion of objective vs. subjective criteria. Objective criteria are data-verifiable (e.g., cost, emissions); subjective criteria rely on expert opinion (e.g., public acceptance).
- **Measurability (w_3)** — how feasibly each criterion can be measured given available data and effort. Based on expert ratings (0–10 scale).
- **Sensitivity (w_6)** — each criterion's elasticity or impact on decision outcomes. Computed via Monte Carlo simulation with random weight perturbations.
- **Cost-Effectiveness (w_7)** — informational benefit of each criterion relative to its assessment cost. Based on expert ratings (0–10 scale).
- **Alignment (w_8)** — how strongly each criterion advances the stated organizational or stakeholder objectives. Based on expert ratings (0–10 scale).
- **Cognitive Coherence (w_9)** — whether stakeholders share a clear, common understanding of each criterion's meaning and scope. Based on cross-expert rating (0–10 scale).

Penalty properties (lower deviation is better):

- **Distinctiveness (w_4)** — penalizes redundancy by measuring pairwise correlations across expert decision matrices. Lower correlation values (closer to 0) indicate more distinct criteria.
- **Parsimony Lower (w_5^-)** — penalizes having too few criteria relative to the minimum bound (ω). Zero penalty when selected criteria $\geq \omega$.
- **Parsimony Upper (w_5^+)** — penalizes having too many criteria relative to the maximum bound (ζ). Zero penalty when selected criteria $\leq \zeta$.
- **Representativeness Min (w_{11}^-)** — penalizes insufficient coverage of decision objectives (shortfall below L_o).
- **Representativeness Max (w_{11}^+)** — penalizes over-concentration on any single objective (excess above U_o).

Reference component confirmation: After selection, a highlighted box shows "Reference: [Component Name] — This component is set to 100 (maximum value)". All other components will be rated relative to this reference (0–100 scale).

Example: If Completeness is selected as the reference, the interface shows "Best: 9.000 $\uparrow$ | Worst: 7.000 $\downarrow$ ". This indicates that Completeness scores range from 7 to 9 in your data, with the swing from worst (7) to best (9) considered the most valuable improvement among all 12 properties.

Note: These definitions correspond directly to the "Objective Breakdown" values shown in your optimization results. When you see contributions like "Completeness: 0.122577" in the results, you can refer back here to understand each component.

4. Rate other components relative to reference

Rating scale: 100 = as valuable as the reference; 50 = half as valuable; 0.0 = not valuable. Adjust sliders for each of the 11 remaining components.

Each slider ranges from 0.0 to 100 in increments of 5, shows the current value above the slider, and includes hover text with a component description.

5. View normalized weights

Right panel — normalized weights display:

- Reference component is highlighted (e.g., "Completeness — Raw 41.67 $\rightarrow$ Normalized 0.0625").

- Other components are sorted by weight, each showing name, code, normalized value, percentage contribution, and a color-coded border.
- Bottom of panel shows "Sum: 1.0000000000" to confirm weights sum to exactly 1.0.

6. Confirmation and navigation

- Green success box: "✅ Weights configured using classical swing weighting! Click 'Next' to run optimization."
- "⬅️ Back" returns to Step 2; "Next ➡️" proceeds to Step 4.

Key differences from direct weighting

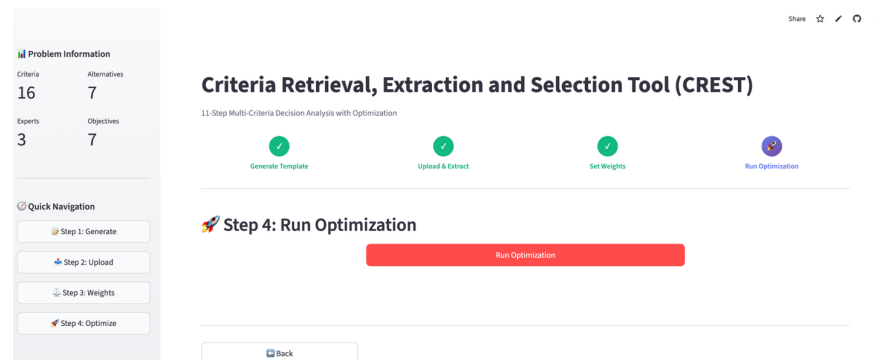
- **Reference-based rating:** all components are judged relative to a single most-important component.
- **Intuitive scale:** 0–100 where 100 means "equally valuable as the reference".
- **Transparent normalization:** raw values are shown alongside normalized weights.
- **Methodologically sound:** follows Edwards & Barron (1994) SMART/SMARTS methodology.

Tips

- Choose your reference component carefully — it should represent the quality aspect you find most critical.
- A rating of 50 means "half as important as the reference" — a common starting point.
- Normalized weights automatically adjust as you change ratings.
- All 12 components receive non-zero weights even if you rate some at 0.0 (due to normalization).

Step 4: Run Optimization

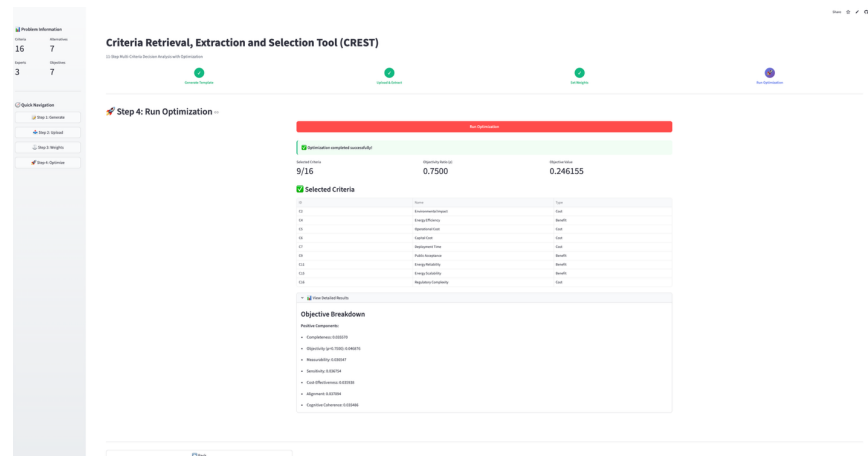
4.1 Execute optimization



Instructions:

42. **Access Step 4.** Progress indicator shows Generate Template ✓ | Upload & Extract ✓ | Set Weights ✓ | Run Optimization (active). Sidebar shows "🚀 Step 4: Optimize" highlighted.
43. **Initiate optimization.** Click the large red "Run Optimization" button. Wait 10–60 seconds while the system builds the optimization model, solves for the optimal criteria subset, and calculates performance metrics.

4.2 View and interpret results



After optimization completes, the page displays comprehensive results.

Success confirmation: Green box "✅ Optimization completed successfully!"

Summary metrics (three columns at top):

44. **Selected Criteria: 9/16.** 9 criteria selected out of 16 total. Falls within parsimony bounds [5, 9]. Represents a parsimonious, manageable set. Range: $\omega \leq \text{selected} \leq \zeta$; smaller, well-justified sets are generally preferred.
45. **Objectivity Ratio (ρ): 0.7500.** 75% of selected criteria are objective. Range: 0.0 to 1.0; higher ρ means a more data-verifiable, less judgment-driven set, but a balanced mix (typically 0.5–0.9) is usually preferable so important subjective factors are not lost.
46. **Objective Value: 0.246155.** Overall optimization score. Range: typically 0 to ~1 depending on weights; higher is better. Use it to compare different weight configurations or constraint settings on the same problem — absolute values are not directly comparable across different problems.

Selected Criteria table: Lists each selected criterion with ID, Name, and Type (Cost/Benefit). Example output: C2 Environmental Impact (Cost), C4 Energy Efficiency (Benefit), C5 Operational Cost (Cost), C6 Capital Cost (Cost), C7 Deployment Time (Cost), C8 Public Acceptance (Benefit), C11 Energy Reliability (Benefit), C15 Energy Scalability (Benefit), C18 Regulatory Complexity (Cost).

Interpretation:

- These are the optimal criteria for your decision problem.
- Mix of Cost and Benefit criteria covering multiple strategic objectives.
- Ready to use in your downstream MCDM analysis.
- Excluded criteria are reported alongside the selected set.

Detailed results (expandable section): Click "📄 View Detailed Results" to see the Objective Breakdown of positive components — for example: Completeness 0.031519, Objectivity ($\rho=0.7500$) 0.046676, Measurability 0.054547, Sensitivity 0.036254, Cost-Effectiveness 0.030339, Alignment 0.037094, Cognitive Coherence 0.019449.

What this tells you: Each value shows that component's contribution to the final selection. Higher values mean greater influence. Compare values to your weight settings in Step 3 to understand the selection drivers.

If results are unexpected

Too many criteria selected (more than ζ): return to Phase 2, Step 3 and increase the Parsimony Upper weight; or reduce ζ in Phase 2, Step 1.

Too few criteria selected (fewer than ω): return to Phase 2, Step 3 and increase the Parsimony Lower weight; or increase ω in Phase 2, Step 1.

Important criterion missing: review its scores in the Excel template across all 11 sheets, check if it is highly correlated with a selected criterion, and consider relaxing relevant thresholds in Phase 2, Step 1.

Run multiple scenarios: try different weight configurations in Phase 2, Step 3, compare across scenarios, and identify criteria that consistently appear (robust selections).

Tips for Successful CREST Implementation

Before you start

- Clearly define your decision problem and objectives.
- Identify all potentially relevant criteria — don't pre-filter.
- Recruit 3–7 qualified experts with diverse perspectives. The framework also runs with a single expert when necessary; in that case, expert-disagreement metrics (e.g., Cognitive Coherence) become less informative and should be interpreted accordingly.
- Allocate sufficient time for thoughtful data collection.

Phase 1: Criteria extraction

- Provide detailed domain context in Step 2.
- Complete at least one optional step (3, 4, or 5).
- Combine multiple sources for comprehensive coverage.
- Review carefully and verify AI-generated criteria align with your needs.
- Export the validation form for expert feedback.

Phase 2: Data collection

- Provide clear rating guidelines to experts.
- Use concrete examples to calibrate understanding.
- Allow discussion and iteration among experts.
- Document assumptions for decision matrices.
- Fill all yellow cells; leave green cells untouched.

Setting weights (Phase 2, Step 3)

- Choose your reference component carefully.
- Start with equal or default weights if uncertain.
- Adjust sliders based on your decision context.
- Watch the normalized weights panel update in real time.
- Verify the sum equals 1.0 at the bottom.
- Test multiple weight configurations for sensitivity.

Interpreting results (Phase 2, Step 4)

- Validate results with domain experts.
- Check the objectivity ratio (e.g., 0.8750 means 87.5% objective criteria).
- Review the selected criteria table for face validity.
- Examine the objective breakdown for contribution patterns.
- Ensure the selected set is practical to implement.

Troubleshooting

Phase 1 issues

I. No criteria generated

Causes: Step 2 not completed, objectives not defined.

Solutions:

- 47. Return to Step 2 and complete domain setup.
- 48. Ensure decision objectives are clearly listed.
- 49. Verify the API key is configured in Step 1.

II. PDF extraction failed

Causes: Scanned images, non-text PDFs, corrupted files.

Solutions:

- 50. Ensure PDFs contain selectable text (not scanned images).
- 51. Try re-exporting PDFs from the original source.
- 52. Check the Analysis Log for specific error messages.

III. Low AI confidence scores

Causes: Unclear domain description, ambiguous objectives.

Solutions:

- 53. Review domain description clarity in Step 2.
- 54. Add more context about constraints and requirements.
- 55. Make objectives more specific and measurable.

IV. Export not downloading

Causes: Browser settings, pop-up blockers.

Solutions:

- 56. Check browser download settings and permissions.
- 57. Disable pop-up blockers for the site.
- 58. Try a different browser.

Phase 2 issues

V. "No optimal solution found"

Causes: Thresholds too strict, impossible constraints, conflicting requirements.

Solutions:

- 59. Return to Phase 2, Step 1 and reduce quality thresholds:
 - Decrease Completeness (α) by 0.5–1.0.
 - Decrease Measurability thresholds (γ_O , γ_S).
 - Increase Distinctiveness (δ) to allow more correlation.
 - Decrease the Sensitivity (θ) threshold.
- 60. Widen parsimony bounds — lower ω , raise ζ .

61. Check that expert ratings allow some criteria to meet thresholds.

VI. Upload/extraction fails (Phase 2, Step 2)

Causes: Wrong file format, file size, missing sheets, blank cells.

Solutions:

62. Verify file format is .xlsx (limit 200 MB per file).
63. Check file size is under 200 MB.
64. Ensure all 12 sheets are present in the Excel file.
65. Verify all yellow cells are filled (no blanks).
66. Use only the generated template — don't copy to a new file.
67. Check for special characters in criteria/alternative names.

VII. Unexpected results (Phase 2, Step 4)

Causes: Weight configuration, inconsistent expert ratings, data quality issues.

Solutions:

68. Review weight configuration in Phase 2, Step 3 — check that sliders reflect your priorities and that normalized weights are reasonable.
69. Check expert ratings for consistency in the Excel file — look for outliers or data-entry errors and verify all experts understood the rating scales.
70. Examine selected vs. excluded criteria — review why excluded criteria scored lower and check for correlation patterns.
71. Run sensitivity analysis with different weights — test multiple scenarios and identify robust selections.

Exporting Your Results

From Phase 1

- 72. **Export CSV/JSON:** contains all criteria with metadata.
- 73. **Generate Report:** comprehensive text summary.
- 74. **Create Validation Form:** template for expert review.
- 75. **Screenshot:** capture the consolidation summary.

From Phase 2

- 76. **Screenshot results:** capture Phase 2 / Step 4 showing all metrics and selected criteria.
- 77. **Copy criteria table:** select and copy the selected criteria with their names and types.
- 78. **Document settings:** record the weight configuration from Phase 2, Step 3 and the parameter settings from Phase 2, Step 1.
- 79. **Save Excel file:** keep your completed template for reference.

Contact and Support

Application URLs:

- **Phase 1:** [Open Phase 1](#)
- **Phase 2:** [Open Phase 2](#)

GitHub repository: [parisahjb/mcdm-dashboard](https://github.com/parisahjb/mcdm-dashboard)

For questions or issues: submit via GitHub Issues.

References

- Edwards, W., & Barron, F. H. (1994). SMARTS and SMARTER: Improved simple methods for multiattribute utility measurement. *Organizational Behavior and Human Decision Processes*, 60(3), 306–325.
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- Keeney, R. L., & Raiffa, H. (1976). *Decisions with multiple objectives: Preferences and value tradeoffs*. New York: John Wiley & Sons.